

SQL Retail Analytics Project Test

Total marks 50

Business task and connection

You are an analyst for TechMart Pakistan, a fictional electronics retailer. Use 2025 sales to identify profitable products, compare stores and recognise valuable customers. Complete five SQL queries and a short management recommendation.

Open retail_project.db. It contains seven tables and 160,034 rows.

Q1 Find online orders

6 marks

From orders, show 10 Delivered Website orders from November 2025. Return order_id, customer_id, order_date and channel. Sort order_date descending, then order_id descending; use LIMIT.

Q2 Typecast and create a column

8 marks

From order_items, show the first 10 lines by order_item_id. Return order_item_id, quantity_text and unit_price_text, plus quantity_num cast to INTEGER, unit_price_num cast to FLOAT, and gross_line_amount calculated by multiplying the two cast values. Repeat the CAST expressions inside the multiplication. This amount is before discounts and returns.

Q3 Identify valuable customers

8 marks

Join customers to orders using customer_id and include only Delivered orders. Show the top five customers by total net revenue. Return customer_id, customer_full_name, order_count and total_net_revenue. Concatenate first_name and last_name with one space. Group at customer level; sort total_net_revenue descending then customer_id ascending; limit to five.

Q4 Rank products with multiple joins

12 marks

In one query join order_items, orders, products, categories and suppliers. Include only Delivered orders. Return the five products with highest total gross profit, showing product_id, product_name, category_name, supplier_name, total_net_units, total_net_revenue and total_gross_profit. Use SUM and GROUP BY; sort gross profit descending then product_id ascending; limit to five.

- order_items.order_id → orders.order_id
- order_items.product_id → products.product_id
- products.category_id → categories.category_id
- products.supplier_id → suppliers.supplier_id

Table guide

customers holds customer names and IDs. orders holds order dates, status, customer_id, store_id and order-level net_revenue and gross_profit. order_items holds order_id, product_id, the numeric text fields, net_units_sold and line-level net_revenue and gross_profit. products links to categories and suppliers. stores holds store names and IDs. See schema for all fields.

Use orders for customer counts and average order value. Use order_items for product totals.

Q5 Compare store performance

10 marks

Join stores to orders using store_id and include only Delivered orders. Return store_id, store_name, delivered_order_count, total_net_revenue, total_gross_profit, average_order_value and profit_margin_pct. Use COUNT, SUM, AVG and GROUP BY. Average order value is $\text{AVG}(\text{orders.net_revenue})$.

Calculate $\text{profit_margin_pct} = \text{gross_profit} / \text{net_revenue} \times 100$. Sort total_gross_profit descending, then store_id ascending.

Q6 Give management a recommendation

6 marks

- Recommend one product to promote and support it with two Q4 values.
- Choose one store for attention, state an action and support it with two Q5 values.
- Suggest one action for valuable customers using Q3 and explain one limitation. For example, store 12 opened only in December 2025 and gross profit excludes operating costs.